Team Bacteria: Talking and Working Together

How do single-celled bacteria cooperate — and can we use that against them?

Bacteria are single cells, but they don't act alone. They build communities, communicate with chemical signals, and launch coordinated attacks. Understanding how they work as a team could be the key to beating superbugs.

PART A — Biofilms: Bacteria's Fortress

Read: *When bacteria stick to a surface and multiply, they can form a* ***biofilm*** *— a slimy layer of bacteria glued together with a protective coating they produce themselves. The plaque on your teeth in the morning is a biofilm. So is the slippery layer on a wet rock. Biofilms can make bacteria up to 1,000 times harder to kill with antibiotics than free-floating bacteria. That's because the outer layers shield the inner bacteria from the drug, and bacteria in biofilms can share resistance genes with each other.*

1. What is a biofilm? Give one real-world example from the reading and one of your own.

2. Why are bacteria in a biofilm harder to kill with antibiotics than bacteria floating freely? Give TWO reasons.

3. Connect to cell transport: how might bacteria in the outer layer of a biofilm use active transport to help bacteria in the inner layers survive antibiotics?

PART B — Quorum Sensing: How Bacteria "Talk"

Read: *Each bacterium constantly releases tiny chemical signal molecules into its surroundings. The more bacteria are present, the stronger the chemical signal gets. When the signal reaches a threshold — a* ***quorum*** *(like enough players to start a game) — ALL the bacteria respond at once. This is called* ***quorum sensing****. Harmful bacteria use it to wait until they have enough numbers before launching an infection — so the immune system is overwhelmed all at once instead of having time to fight back.*

4. What is quorum sensing? Explain it in your own words as if you're explaining it to a younger student.

5. Why is it an advantage for harmful bacteria to WAIT until quorum is reached before launching an infection? Think about what the immune system does.

6. Analogy: Dr. Bonnie Bassler, the scientist who discovered quorum sensing, compared bacteria to tiny spies who "count" each other. Think of a sports analogy — how is quorum sensing like something that happens in volleyball?

PART C New Weapons: Disrupting Bacterial Teamwork

Big Idea: *Instead of killing bacteria directly (which creates selection pressure for resistance), scientists like Dr. Bassler are designing drugs that BLOCK quorum sensing signals. Without communication, bacteria can't coordinate an attack. They can't form biofilms. And because the drug doesn't kill them — just confuses them — there's less pressure to evolve resistance.*

7. Why might blocking quorum sensing create LESS antibiotic resistance than killing bacteria directly? Use what you know about natural selection.

8. Could this strategy work against a bacterium that has already become resistant to antibiotics? Explain your reasoning.

PART D Cooperation in Nature — Beyond Bacteria

Bacteria aren't the only organisms that use teamwork. Match each example of cooperative behavior to what it has in common with quorum sensing or biofilms.

Example	Like biofilm or quorum sensing?	What they have in common
Humpback whales bubble-net fishing		
A volleyball team waiting for the setter's signal before attacking		
Hawaiian bobtail squid hosting bioluminescent bacteria		
Ants carrying food together that no single ant could move		

■ **Exit Ticket** Imagine you are a scientist designing a new drug to fight biofilm infections. Would you try to: (a) kill all bacteria, (b) block quorum sensing, or (c) break up the biofilm? Choose ONE and defend your choice using evidence from today's activity.

Adapted for 7th grade from New Visions for Public Schools, Humans vs. Bacteria (Biology Unit 2), CC BY-NC-SA 4.0. Video: Bonnie Bassler — How Bacteria Talk (TED) · Unit page: yscabarney.github.io/MS_SCI_ADV_26/humans-vs-bacteria.html